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Building Responsible AI

Innovation, Ethics and Security

AI: promise vs responsibility

AI is Transforming the World — But Can We Guide It Responsibly?

$1.8 trillion

Global AI market by 2030

- Technical overview
- Practical Applications
- Ethical Challenges
- Security Risks
- Regulatory Frameworks
- Case Studies

How AI Thinks

Foundation of AI

Machine Learning

For pattern recognition
(Mazzei and Ramjattan, 2022)

Deep Learning

For complex data processing
(LeCun, Bengio and Hinton, 2015)

For human-AI interaction
(Min et al., 2023)

Natural Language Processing

For image and video analysis
(Zhao et al., 2019)

Computer vision

(Russell & Norvig, 2020)

AI in action

Healthcare

DeepMind's AKI prediction system reduces missed cases from 12.4% to 3.3% with 48-hour early detection capabilities
(Google DeepMind, 2018)

Financial

JPMorgan's IndexGPT leverages GPT-4 to generate superior investment themes with twice the keyword coverage
(Sai et al., 2025)

Enterprise

AI-powered chatbots provide 24/7 customer service and automate routine tasks

(Følstad & Brandtzaeg, 2017)

HR Dpt.

AI-driven recruitment tools screen CVs at scale, with 99% of Fortune 500 companies automating early hiring stages

(Milne, 2024)

Ethical dilemmas

Algorithmic bias

AI systems perpetuate historical biases from training data

(Marieke and Sach, 2024)

Privacy violations

AI systems often rely on personal data, raising concerns about consent, and misuse

(Cadwalladr & Graham-Harrison, 2018)

Lack of transparency

AI models often operate as 'black boxes', making their decisions difficult to interpret

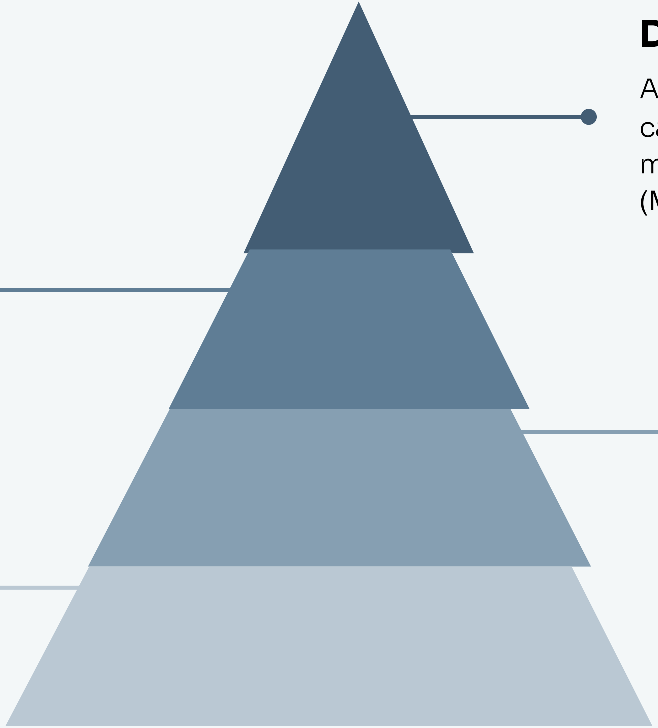
(Rudin, 2019)

Job displacement

AI automation is replacing human roles, raising concerns about unemployment

(Bonney et al., 2024)

Security threats



Deepfakes and Misinformation

AI-generated media like deepfakes can spread disinformation and manipulate public perception (Momeni, 2024)

Model Theft

Attackers can replicate AI models by querying them repeatedly, threatening intellectual property (Liang et al., 2024)

Adversarial Attacks

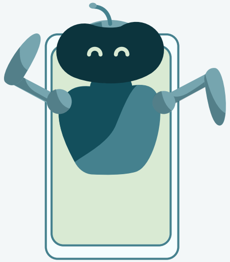
Small, intentional changes to input data can mislead AI into making incorrect predictions (Girdhar, Hong and Moore, 2023)

Data Poisoning

Malicious data can be injected into training sets, corrupting AI systems from the inside (Yao et al., 2024)

Governing the algorithms

Framework	Focus Area	Key Principle
GDPR	Data privacy + explainability	Right to explanation (Ljubiša Metikoš and Ausloos, 2025)
EU AI Act	Risk-based system classification	Human oversight, transparency (Schuett, 2022)
NIST AI RMF	Voluntary U.S. risk management	Trustworthiness, fairness, security (NIST, 2023)



From headlines to lessons

Facebook's proactive deepfake detection challenge

Facebook Response

- Launched global Deepfake Detection Challenge in 2020
- Provided large benchmark dataset for researchers
- Emphasized AI-powered tools with human oversight

(Meta, 2020)

Amazon's biased AI recruiting tool failure

Amazon Lessons

- Discontinued biased AI recruiting tool in 2018
- System discriminated against female applicants systematically
- Highlighted need for bias audits and diversity

(Dastin, 2018)

Microsoft's chatbot manipulation vulnerability

Microsoft Experience

- Tay chatbot corrupted within 24 hours
- Coordinated trolling generated offensive content
- Demonstrated AI vulnerability to adversarial input

(Brooker, 2019)

The path forward

01 Ethics and security must evolve alongside innovation

02 Governance, transparency, and fairness must be built into AI systems

03 Case studies show why proactive regulation and oversight are essential

Thank you for your attention!

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